

Chapter 6 — Interest Rate Futures

J.C. Hull — *Options, Futures and Other Derivatives*

6.1 Introduction to Interest Rate Futures

Interest rate futures are futures contracts where the underlying asset depends on interest rates. These contracts are among the most widely traded financial derivatives in the world because interest rates affect:

- Bond prices
- Borrowing costs
- Lending rates
- Bank profitability
- Corporate financing
- Portfolio values

Earlier chapters discussed:

- Commodity futures
- Stock index futures
- Currency futures

This chapter focuses specifically on **interest rate futures**.

The two most important contracts discussed are:

1. **Treasury Bond Futures**
2. **Eurodollar Futures**

These contracts are heavily used for:

- Hedging interest rate risk
 - Speculation on future rates
 - Portfolio duration management
 - Arbitrage trading
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Why Interest Rate Futures Matter

Interest rates are one of the most important variables in finance.

A small movement in rates can create very large changes in:

- Bond values
- Loan costs
- Bank assets/liabilities
- Fixed income portfolios

Interest rate futures allow market participants to:

- Lock in future borrowing costs
 - Lock in future lending returns
 - Hedge bond portfolios
 - Speculate on Federal Reserve policy
 - Manage duration exposure
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Relationship Between Interest Rates and Bond Prices

There is an inverse relationship:

- When interest rates rise → bond prices fall
- When interest rates fall → bond prices rise

This inverse relationship is the foundation of interest rate futures trading.

6.1 Day Count Conventions

Before understanding interest rate futures, Hull first explains **day count conventions** because interest calculations in bond markets depend on them.

A day count convention determines:

1. How days between dates are counted

2. How interest accrues over time

General notation:

$$X/Y$$

Where:

- X = method for counting days between dates
 - Y = method for measuring days in year/reference period
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Formula for Accrued Interest

$$\text{Interest Earned} = \frac{\text{Days Elapsed}}{\text{Days in Reference Period}} \times \text{Coupon Payment}$$

Major Day Count Conventions

1. Actual/Actual (in period)

Used for:

- US Treasury Bonds

Characteristics:

- Uses actual calendar days
 - Most accurate method
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Example — Actual/Actual Convention

Suppose:

- Bond principal = \$100
- Coupon rate = 8% annually
- Coupons paid semiannually

Therefore:

- Coupon every 6 months = \$4

Coupon dates:

- March 1
- September 1

We calculate accrued interest between:

- March 1 and July 3

Step 1: Count actual days

Days from March 1 to July 3:

124 days

Days between coupon dates:

184 days

Step 2: Compute accrued interest

$$\frac{124}{184} \times 4 = 2.6957$$

Thus accrued interest:

\$2.70

2. 30/360 Convention

Used for:

- Corporate bonds
- Municipal bonds

Assumptions:

- Every month = 30 days
 - Every year = 360 days
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Example — 30/360

March 1 to September 1:

180 days

March 1 to July 3:

$$(4 \times 30) + 2 = 122 \text{ days}$$

Accrued interest:

$$\frac{122}{180} \times 4 = 2.7111$$

Thus:

\$2.71

3. Actual/360 Convention

Used for:

- Money market instruments
- LIBOR calculations

Characteristics:

- Actual days counted
- Year assumed to have 360 days

Interest for part of year:

$$\text{Interest} = \text{Rate} \times \frac{\text{Actual Days}}{360}$$

Important Observation

Different conventions produce different interest amounts.

This is extremely important in:

- Bond pricing
 - Futures pricing
 - Swap valuation
 - LIBOR calculations
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Business Snapshot — Day Counts Can Be Deceptive

Hull points out an interesting fact.

Suppose:

- Treasury bond
- Corporate bond

Both:

- Same coupon
- Same quoted price

Between Feb 28 and March 1:

- Treasury bond uses actual/actual → 1 day
- Corporate bond uses 30/360 → 3 days

Therefore:

- Corporate bond earns nearly 3× more accrued interest during that period.

This demonstrates how conventions affect pricing and valuation.

International Day Count Conventions

Different countries use different standards.

Examples:

Instrument	Convention
US Treasury Bonds	Actual/Actual
US Corporate Bonds	30/360
LIBOR	Actual/360
Sterling LIBOR	Actual/365
Euro Bonds	Actual/Actual

Treasury Bill Price Quotations

US Treasury bills use a special quotation system based on a **discount rate**.

The quoted rate is:

- Interest as percentage of face value
- Not percentage of purchase price

Treasury Bill Pricing Formula

$$P = \frac{360}{n}(100 - Y)$$

Where:

- P = quoted price
- Y = cash price
- n = days remaining

Example — Treasury Bill

91-day Treasury bill quoted at:

Meaning:

- 8% discount rate on face value

Face value:

\$100

Interest earned:

$$100 \times 0.08 \times \frac{91}{360} = 2.0222$$

True return:

$$\frac{2.0222}{100 - 2.0222} = 2.064\%$$

for the 91-day period.

Price Quotations of US Treasury Bonds

Treasury bonds are quoted in:

- Dollars
- Thirty-seconds of a dollar

Example:

90 – 05

means:

$$90 + \frac{5}{32} = 90.15625$$

For \$100,000 face value:

\$90,156.25

Clean Price vs Dirty Price

Two important bond prices:

1. Clean Price

- Quoted market price
- Excludes accrued interest

2. Dirty Price (Cash Price)

- Actual amount paid
- Includes accrued interest

Formula:

$$\text{Cash Price} = \text{Quoted Price} + \text{Accrued Interest}$$

Example — Dirty Price

Bond:

- 11% coupon
- Maturity: July 10, 2038
- Quoted price: 95-16

Convert quote:

$$95 + \frac{16}{32} = 95.50$$

Coupon payment:

$$\frac{11}{2} = 5.50$$

Days:

- Since last coupon = 54
- Between coupon dates = 181

Accrued interest:

$$\frac{54}{181} \times 5.50 = 1.64$$

Cash price:

$$95.50 + 1.64 = 97.14$$

For \$100,000 face value:

\$97,140

6.2 Treasury Bond Futures

Treasury bond futures are among the most important interest rate futures contracts.

Traded mainly on:

- CME Group

Underlying asset:

- US government bonds
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Deliverable Bonds

In Treasury bond futures:

- Any government bond with maturity between 15 and 25 years may be delivered.

Ultra T-bond futures:

- Deliverable bonds have maturity greater than 25 years.
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Treasury Note Futures

Other popular contracts:

Contract	Deliverable Maturity
10-Year Note	6.5–10 years
5-Year Note	About 5 years

Contract	Deliverable Maturity
2-Year Note	About 2 years

Treasury Futures Quotes

Treasury futures are quoted similarly to Treasury bonds.

Example:

$$144 - 20$$

means:

$$144 + \frac{20}{32} = 144.625$$

Precision in Treasury Note Quotes

10-year notes:

- Quoted to nearest half of 1/32

5-year and 2-year notes:

- Quoted to nearest quarter of 1/32

This allows highly precise pricing.

Conversion Factors

The short party can choose among many deliverable bonds.

To standardize delivery, the exchange uses a:

Conversion Factor

Purpose of Conversion Factor

Different bonds have:

- Different coupons
- Different maturities

Conversion factors adjust for these differences.

Invoice Price Formula

Amount received by short:

$$(\text{Futures Settlement Price} \times \text{Conversion Factor}) + \text{Accrued Interest}$$

Example — Invoice Price

Suppose:

- Futures price = 90-00
- Conversion factor = 1.3800
- Accrued interest = \$3

Invoice price:

$$(90 \times 1.38) + 3 = 127.20$$

For \$100,000 face value:

\$127,200

How Conversion Factors Are Calculated

The exchange assumes:

- Yield = 6% per annum
- Semiannual compounding

The conversion factor equals:

- Theoretical bond price per \$1 principal under 6% yield assumption
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Important Rules

When computing conversion factor:

- Maturity rounded down to nearest 3 months
 - Coupon dates adjusted accordingly
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Example — Conversion Factor

10% coupon bond:

- 20 years 2 months maturity

Rounded to:

- 20 years

Discounting at:

- 6% annual yield

Bond value becomes:

146.23

Conversion factor:

1.4623

Cheapest-to-Deliver (CTD) Bond

The short position chooses which bond to deliver.

Naturally, the short chooses:

Cheapest-to-Deliver Bond

CTD Cost Formula

Short receives:

$$(\text{Futures Price} \times \text{Conversion Factor}) + \text{Accrued Interest}$$

Short pays:

- Market purchase price of bond

CTD minimizes:

$$\text{Quoted Bond Price} - (\text{Futures Price} \times \text{Conversion Factor})$$

Example — Cheapest to Deliver

Suppose futures settlement price:

93.25

Bond	Price	Conversion Factor
1	99.50	1.0382
2	143.50	1.5188
3	119.75	1.2615

Bond 1

$$99.50 - (93.25 \times 1.0382) = 2.69$$

Bond 2

$$143.50 - (93.25 \times 1.5188) = 1.87$$

Bond 3

$$119.75 - (93.25 \times 1.2615) = 2.12$$

Lowest cost:

- Bond 2

Therefore:

Bond 2 is Cheapest-to-Deliver

Factors Affecting CTD Bond

If yields > 6%

- Low coupon
- Long maturity

tend to be favored.

If yields < 6%

- High coupon
- Short maturity

tend to be favored.

Yield Curve Effects

Upward Sloping Yield Curve

- Long maturity bonds favored

Downward Sloping Yield Curve

- Short maturity bonds favored
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Wild Card Play

Treasury bond futures contain embedded options for the short.

One important option:

Wild Card Play

Why It Exists

Settlement price fixed at:

- 2:00 PM Chicago time

But Treasury bonds continue trading afterward.

The short may:

- Observe bond market after 2 PM
 - Decide whether to issue delivery notice later
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How Wild Card Play Works

If bond prices fall after 2 PM:

- Short buys bonds cheaply in spot market
- Delivers using earlier futures settlement price

This creates profit opportunity.

Important Insight

The short position has valuable delivery options:

- Choice option
- Timing option
- Wild card option

These reduce the fair futures price.

Determining Treasury Bond Futures Price

Exact pricing is difficult because:

- Short has delivery options

But assuming:

- Delivery date known
- CTD bond known

Then futures pricing resembles pricing of an asset with known income.

Treasury Bond Futures Pricing Formula

$$F_0 = (S_0 - I)e^{rT}$$

Where:

- F_0 = futures price
 - S_0 = spot price
 - I = present value of coupons during contract life
 - r = risk-free rate
 - T = time to maturity
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Example — Futures Pricing

Suppose:

- Bond coupon = 12%
 - Conversion factor = 1.6000
 - Delivery in 270 days
 - Rate = 10% continuously compounded
 - Bond quoted price = \$115
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Step 1 — Find Cash Price

$$115 + \frac{60}{60 + 122} \times 6 = 116.978$$

Step 2 — PV of Coupon

$$6e^{-0.1 \times 0.3342} = 5.803$$

Step 3 — Futures Cash Price

$$(116.978 - 5.803)e^{0.1 \times 0.7397} = 119.711$$

Step 4 — Convert to Quoted Futures Price

After accrued interest adjustment:

114.859

Using conversion factor:

$$\frac{114.859}{1.6000} = 71.79$$

Thus quoted futures price:

71.79

6.3 Eurodollar Futures

Eurodollar futures are:

- Futures on short-term interest rates

Most actively traded interest rate futures globally.

What is a Eurodollar?

A Eurodollar is:

- US dollar deposit
- Held outside United States

Eurodollar rate is essentially:

LIBOR

Eurodollar Futures Contract

Underlying:

- Interest on \$1 million deposited for 3 months

Used for:

- Hedging short-term rates
 - Speculation on future LIBOR
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Contract Maturities

Contracts available:

- March
- June
- September
- December

Up to:

- 10 years ahead
-

Eurodollar Futures Quote Convention

Price quote:

$$100 - R$$

Where:

- R = forward LIBOR interest rate (%)
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Example

If futures quote:

99.725

Then implied LIBOR:

$$100 - 99.725 = 0.275\%$$

Relationship Between Eurodollar Futures and Rates

- Rates rise → futures price falls
- Rates fall → futures price rises

Same inverse relationship as bonds.

Hedging with Interest Rate Futures

Interest rate futures are widely used to hedge:

- Bond portfolios
 - Interest-sensitive assets/liabilities
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Key Principle

When interest rates rise:

- Futures prices fall

When interest rates fall:

- Futures prices rise

Hedging Rules

If portfolio loses when rates rise:

→ Take SHORT futures position

If portfolio loses when rates fall:

→ Take LONG futures position

Duration-Based Hedging

The key tool:

Duration

Duration measures:

- Sensitivity of bond price to interest rates
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Hedge Ratio Formula

$$N = \frac{PD_P}{FD_F}$$

Where:

- N = number of futures contracts
- P = portfolio value
- D_P = portfolio duration
- F = futures contract value
- D_F = duration of CTD bond

Example — Duration Hedge

Portfolio:

- \$10 million

Portfolio duration:

- 6.8 years

Futures price:

- 93-02 = 93.0625

Contract size:

93,062.50

CTD duration:

- 9.2 years

Calculate Contracts

$$N = \frac{10,000,000 \times 6.8}{93,062.5 \times 9.2} = 79.42$$

Thus:

Short 79 contracts

Portfolio Immunization

Financial institutions often try to match:

- Asset duration
- Liability duration

This is called:

Duration Matching

or

Portfolio Immunization

Goal of Immunization

Protect against:

- Small parallel shifts in interest rates
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Limitation of Duration Matching

It works poorly when:

- Yield curve shifts are nonparallel

Example:

- Short-term rates rise sharply
- Long-term rates move slightly

Then hedge becomes imperfect.

Key Exam Concepts from Chapter 6

You must know:

Concepts

- Day count conventions
 - Clean vs dirty price
 - Treasury futures quotation
 - Conversion factors
 - Cheapest-to-deliver bond
 - Wild card play
 - Eurodollar futures
 - Duration hedging
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Important Formulas Summary

Accrued Interest

$$\frac{\text{Days Elapsed}}{\text{Days in Period}} \times \text{Coupon}$$

Dirty Price

$$\text{Cash Price} = \text{Quoted Price} + \text{Accrued Interest}$$

Treasury Futures Pricing

$$F_0 = (S_0 - I)e^{rT}$$

Hedge Ratio

$$N = \frac{PD_P}{FD_F}$$

Final Intuition of Chapter 6

This chapter connects:

- Bond pricing
- Interest rates
- Futures markets
- Duration risk

Core intuition:

- Bond prices and interest rates move oppositely.
- Interest rate futures allow traders to transfer interest rate risk.
- Treasury futures involve delivery complications.
- Duration is central to hedging fixed-income portfolios.
- Eurodollar futures dominate short-term rate trading globally.